

Demo Stylish Article

Félix Romiguière ¹

Résumé

This document is only a demo explaining how to use the Stylish Article template.

Keywords

template, demo

¹Université de Montpellier, Scientific Department, Somewhere, City, Country, 9999

Table des matières

1	Introduction	1
2	Matériels et Méthodes	1
3	Résultats	2
4	Discussion	2
	References	2

1. Introduction

La question de la diversité des forêts tropicales fascine les écologues parce qu'elle a des enjeux très importants (Gibson et al. 2011). est qu'elle est difficile à comprendre (Wright 2002).

A l'heure des menaces sur la biodiversité (Fadrique et al. 2026), une connaissance plus détaillée de la diversité des forêts très étudiées est utile.

Des modèles ont été publiés (Liang et al. 2022).

Comparer la diversité de Paracou (Gourlet-Fleury et al. 2004) et de BCI (Raby 2017).

2. Matériels et Méthodes

Nombre de Hill (Hill 1973)

- Site de Paracou Comparer la diversité de Paracou (Gourlet-Fleury et al. 2004) et de BCI (Raby 2017).
- Site de BCI
- Les analyses sont réalisées avec R (R Core Team 2026) et le package divient (Marcon et Puech under review).
- Site Selection

The sites and data used in this study comes from the work of Emma Pilleboue, who installed temporary acoustic recording unit (ARU) throughout her internship. After installation, the ARU recorded continuously during a period of several days varying between one full day and five full days, between the 15/02/2024 and 02/05/2024.

In totals, 51 sites of Emma's work are used for this study. The ARU were located at a minimal distance of approximately 200 meters and were in place following altitudinal or urban gradients in some cases. They were also classified into four main habitat type : Urban (4), Suburban (15), Rural (6), Forest(26). The habitats site are mostly forested landscapes or suburban areas, Urban and rural areas are less represented in the site selection.

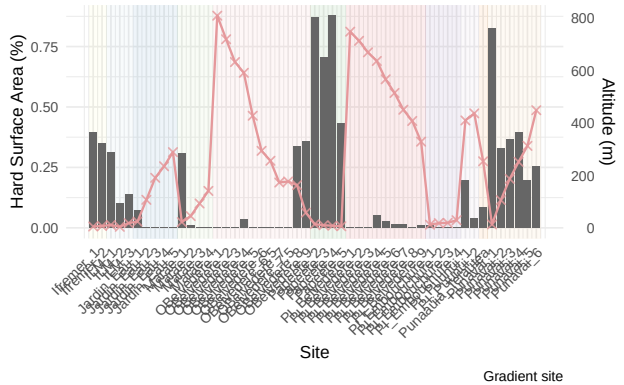
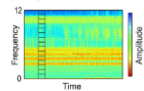
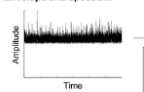
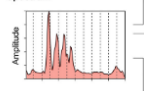
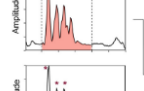
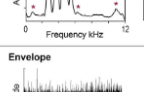
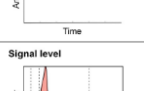
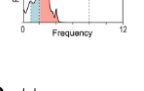
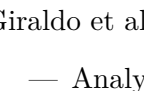


Figure 1. Gradient d'HardSurfaceArea (%) et d'Altitude (m) en fonction des sites étudiés

Acoustic indices selection for better description and tests, use the website to have basic indices described [here](#) (Bradfer-Lawrence et al. 2025)

Complexity	Description	Soundscape pattern	References
Spectrogram 	Acoustic Complexity Index (ACI) Quantifies the sound complexity for estimating the variability of the intensities between time samples within a frequency band. Range=[0,+]	High acoustic index values (↑): - Higher levels of bird activity - Stridulating insects. - Irregular noises from wind, rainfalls.	-Piretti et al., 2011 -Bradler-Lawrence et al., 2019 -Sánchez-Giraldo et al., 2020
Envelope and spectrum 	Acoustic Entropy Index (H) Results from multiplying a temporal diversity index and a frequency diversity index, which are generated by applying Shannon diversity index to the amplitude envelope. Range=[0,1]	- Greater evenness of amplitude among frequency bands and over time. - Soundscape containing many vocalizing species. - Near-silent recordings.	-Sunur et al., 2009b -Bradler-Lawrence et al., 2019
Spectrum 	Acoustic Diversity Index (ADI) Computed by applying the Shannon diversity index to the relative proportion of signal occurring in each 1 kHz frequency band. Range=[0,+]	- Greater evenness soundscapes. - Greater number of occupied frequency. - Rainfall, wind, cicades. - Very quiet recordings	-Villanueva-Rivera et al., 2011 -Bradler-Lawrence et al., 2019 -Sánchez-Giraldo et al., 2020
	Acoustic Evenness Index (AEI) Measures the evenness of the acoustic activity distribution estimating the Gini coefficient on the signal proportion in each 1 kHz band. AEI exhibits an inverse pattern to that of ADI. Range=[0,1]	Low acoustic index values (↓): - Saturated soundscape. - Acoustically rich habitats. - Windy or rainy recordings. - No acoustic activity.	-Villanueva-Rivera et al., 2011 -Bradler-Lawrence et al., 2019 -Sánchez-Giraldo et al., 2020
	Bioacoustic Index (BI) Estimates the area under curve of the mean spectrum above a specific dB threshold within a frequency range. Range=[0,+]	- Higher avian abundance and occupancy of frequency bands. - Blanket cicada noise.	-Boelman et al., 2007 -Bradler-Lawrence et al., 2020
	Number of Peaks (NP) Assesses the complexity by counting the number of major frequency peaks obtained on a mean spectrum. Range=[0,+]	- Habitats containing a higher number of singing species. - Single species producing a noise-like (i.e. cicades).	-Gasc et al., 2013
Envelope 	Median of Amplitude Envelope (M) Estimates the median of amplitude envelope, reflecting the intensity of acoustic recording. Range=[0,+]	- Noisier soundscapes. - Louder recordings. - Recordings with containing rainfalls or winds.	-Depraetere et al., 2012 -Gasc et al., 2015
Signal level 	Normalized Difference Soundscape (NDSI) Calculates the ratio of signal power in the frequency bands between 1-2 kHz (anthrophony) and 2-8 kHz (biophony). Range=[-1,1]	- Higher levels of biophonic activity, and minimal noise in 1-2 kHz band.	-Kasten et al., 2012 -Bradler-Lawrence et al., 2019
	Energy level of Biophony (BIO) Estimates the signal power in the 2-8 kHz frequency band (biophony). Range=[0,+]	- Higher levels of biophonic activity in the soundscape.	-Kasten et al., 2012

Pattern expectation of selected indices (Sánchez-Giraldo et al. 2021)

— Analysis framework intended

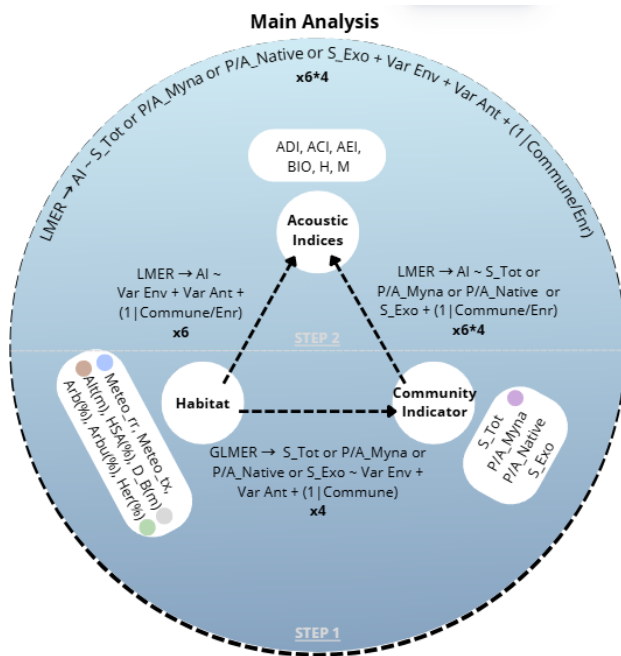


Figure 2. Schéma d'analyse visé

3. Résultats

4. Discussion

MacArthur et Wilson (1967) montrent mathématiquement que la richesse spécifique (le nombre

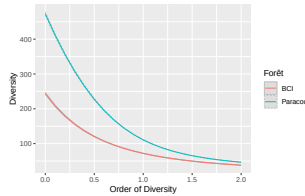


Figure 3. Différence de valeurs de diversité en fonction de son ordre entre les sites de Paracou et BCI

d'espèces) d'une île est un équilibre dynamique entre le taux d'immigration et le taux d'extinction, ce dernier étant directement dicté par la taille de l'île.

BCI est une forêt secondarisée (Raby 2017). Sa diversité pourrait être augmentée selon la théorie de la perturbation intermédiaire (Connell 1978).

Bonus sur la thématique Tahiti_Soundscape

References

Bradler-Lawrence, Tom, Brad Duthie, Carlos Abrahams, et al. 2025. « The Acoustic Index User's Guide : A Practical Manual for Defining, Generating and Understanding Current and Future Acoustic Indices ». *Methods in Ecology and Evolution* 16 (6) : 1040-50. <https://doi.org/10.1111/2041-210X.14357>.

Connell, Joseph H. 1978. « Diversity in tropical rain forests and coral reefs ». *Science* 199 (4335) : 1302-10. <https://doi.org/10.1126/science.199.4335.1302>.

Fadrique, B., F. Costa, F. Cuesta, et al. 2026. « Tree diversity is changing across tropical Andean and Amazonian forests in response to global change ». *Nature Ecology & Evolution* 10 (2) : 267-80. <https://doi.org/10.1038/s41559-025-02956-5>.

Gibson, Luke, Tien Ming Lee, Lian Pin Koh, et al. 2011. « Primary forests are irreplaceable for sustaining tropical biodiversity ». *Nature* 478 (7369) : 378-81. <https://doi.org/10.1038/nature10425>.

Gourlet-Fleury, Sylvie, Jean Marc Guehl, et Olivier Larous-sinie. 2004. *Ecology & management of a neotropical rainforest. Lessons drawn from Paracou, a long-term experimental research site in French Guiana*. Elsevier.

Hill, M. O. 1973. « Diversity and Evenness : A Unifying Notation and Its Consequences ». *Ecology* 54 (2) : 427-32. <https://doi.org/10.2307/1934352>.

Liang, Jingjing, Javier G. P. Gamarra, Nicolas Picard, et al. 2022. « Co-limitation towards lower latitudes shapes global forest diversity gradients ». *Nature Ecology & Evolution* 6 (août) : 1423-37. <https://doi.org/10.1038/s41559-022-01831-x>.

MacArthur, Robert H., et Edward O. Wilson. 1967. *The*

theory of island biogeography. Monographs in population biology. Princeton University Press.

Marcon, Eric, et Florence Puech. under review. « *divent* : an R Package for Diversity Measures Based on Entropy ». *Journal of Open Source Software*, under review.

R Core Team. 2026. *R : A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. <https://doi.org/10.32614/R.manuals>.

Raby, Megan. 2017. *American tropics : the Caribbean roots of biodiversity science*. Flows, migrations, et exchanges. The University of North Carolina Press.

Sánchez-Giraldo, Camilo, Camilo Correa Ayram, et Juan M. Daza. 2021. « Environmental sound as a mirror of landscape ecological integrity in monitoring programs ». *Perspectives in Ecology and Conservation* 19 (3) : 319-28. <https://doi.org/10.1016/j.pecon.2021.04.003>.

Wright, S. Joseph. 2002. « Plant diversity in tropical forests : a review of mechanisms of species coexistence ». *Oecologia* 130 (1) : 1-14. <https://doi.org/10.1007/s004420100809>.